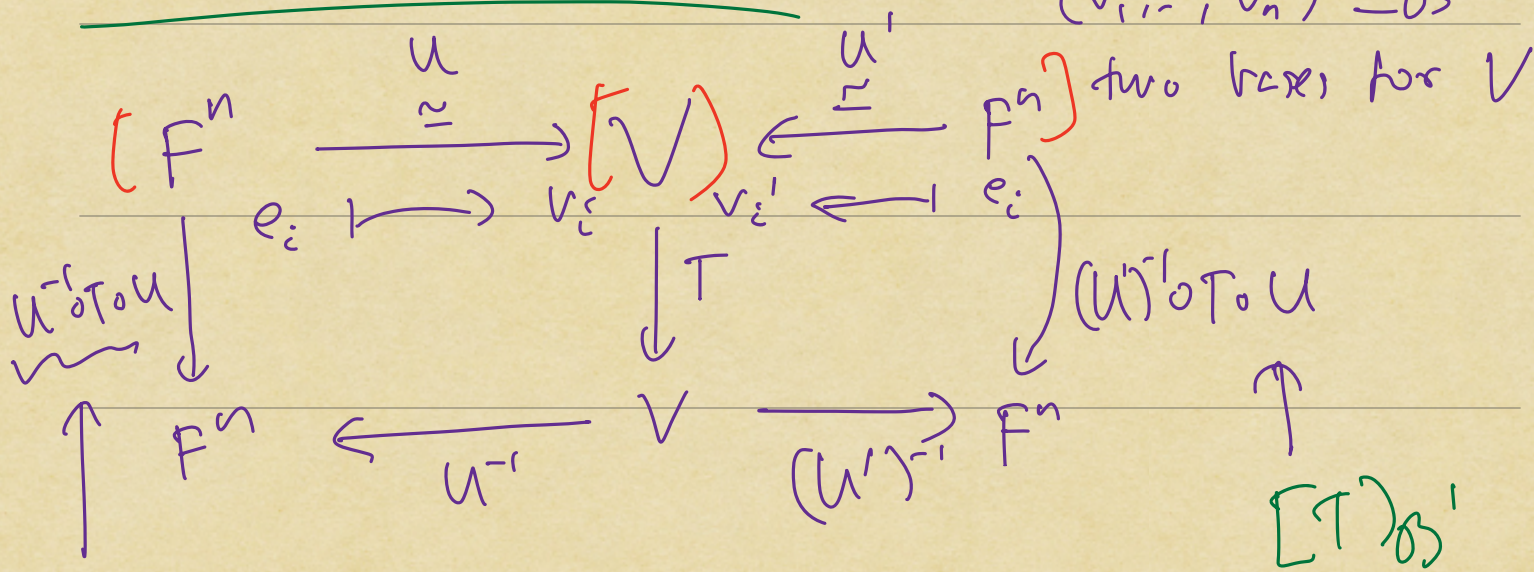


Change of basis II

$$(v_1, \dots, v_n) \subseteq \mathcal{B}$$

$$(v_1', \dots, v_n') = \mathcal{B}'$$

two box, for V



given by

$$[T]_B$$

$$(U^{-1} \circ T \circ U)(e_j) = U^{-1}(T(v_j))$$

$$T(v_j) = \sum_{i=1}^n a_{i,j} v_i$$

then $U^{-1}(T(v_j)) = \sum_{i=1}^n \boxed{a_{i,j}} e_i$

↑
Entries of $[T]_{\beta}$

invertible matrix

$$F^n \xrightarrow{[(U')^{-1} \circ U]} F^n$$

$$\begin{array}{ccc}
 [T]_B & & [T]_{B'} \\
 \downarrow & & \downarrow \\
 F^n & \xleftarrow{[U'^{-1} \circ U']} & F^n \\
 & \uparrow C^{-1} &
 \end{array}$$

$$[T]_B = C^{-1} [T]_{B'} \boxed{C} \leftarrow \text{change of basis matrix}$$

$C \uparrow C^{-1}$

$$C [T]_B C^{-1} = [T]_{B'}$$

What is C?

$$\begin{array}{ccc}
 F^n & \xrightarrow{U} & F^n \\
 \left(\begin{array}{c} e_i \end{array} \right) & \mapsto & \left(\begin{array}{c} v_i \end{array} \right)
 \end{array}$$

$v_i' \xrightarrow{(U')^{-1}} e_i$
 $\searrow$
 F^n

↑
What is this?

Write

$$V_j = \sum_{i=1}^n c_{i,j} V_i'$$

$$\downarrow (U')^{-1}$$

$$(U')^{-1}(V_j) = \sum_{i=1}^n c_{i,j} (U')^{-1}(V_i')$$

$$Ce_j = (U')^{-1}(U(e_j)) = \sum_{i=1}^n c_{i,j} e_i$$

$$Ce_j = \sum_{i=1}^n c_{i,j} e_i$$

Steps: (1) $V_j = \sum_{i=1}^n c_{i,j} V_i'$

(2) $C = [c_{i,j}]$

(3) $[T]_{\mathcal{B}'} = \underbrace{C} [T]_{\mathcal{B}} \underbrace{C^{-1}}$

Example:

$$V = \mathbb{F}^2$$

$$T \mapsto \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = A$$

$$\mathcal{B}' = (e_1, e_2), \quad \mathcal{B} = (\underbrace{v_1, v_2})$$

$$[T]_{\mathcal{B}'} = A \quad ; \quad [T]_{\mathcal{B}} = \begin{bmatrix} 2+\sqrt{3} & 0 \\ 0 & 2-\sqrt{3} \end{bmatrix}$$

Change of basis matrix: $\mathcal{B}$ basis should be expressed in terms $\mathcal{B}'$

$$v_1 = \underbrace{(\sqrt{3}-1)} e_1 + \underbrace{2} e_2$$

$$v_2 = \underbrace{-(\sqrt{3}+1)} e_1 + \underbrace{2} e_2$$

column entries

$$C = \begin{bmatrix} \sqrt{3}-1 & -(\sqrt{3}+1) \\ 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} v_1 & v_2 \end{bmatrix}$$

$$\begin{matrix} [T^m]_{\mathcal{B}'} & = & C [T^m]_{\mathcal{B}} C^{-1} \\ || & & || \end{matrix}$$

$$\begin{matrix} A^m & = & C [T^m]_{\mathcal{B}} C^{-1} \\ || & & || \end{matrix}$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}^m = \begin{bmatrix} \sqrt{3}-1 & -(\sqrt{3}+1) \\ 2 & 2 \end{bmatrix} \begin{bmatrix} (2+\sqrt{3})^m & 0 \\ 0 & (2-\sqrt{3})^m \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3}-1 & -(\sqrt{3}+1) \\ 2 & 2 \end{bmatrix}^{-1}$$

Rank: $\begin{bmatrix} \sqrt{3}-1 & -(\sqrt{3}+1) \\ 2 & 2 \end{bmatrix}^{-1}$ can be found using

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (ad-bc)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Rmk:

$$\begin{array}{ccccc} V & \xrightarrow{T} & V & \xrightarrow{S} & V \\ & \searrow & & \nearrow & \\ & S \circ T & & & \end{array}$$

$$\begin{array}{ccc} [S \circ T]_{\mathcal{B}'} & = & [S]_{\mathcal{B}'} [T]_{\mathcal{B}'} \\ || & & || \end{array}$$

$$C [S \circ T]_{\mathcal{B}} C^{-1} \stackrel{?}{=} \underbrace{(C [S]_{\mathcal{B}} C^{-1}) (C [T]_{\mathcal{B}} C^{-1})}$$

$$= C [S]_{\mathcal{B}} \underbrace{(C^{-1} C)}_{I_n} [T]_{\mathcal{B}} C^{-1}$$

$$= C [S]_{\mathcal{B}} [T]_{\mathcal{B}} C^{-1}$$

$$= C [S \circ T]_{\mathcal{B}} C^{-1}.$$

Properties of 2x2-determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$(1) \det(I_2) = 1$$

$$(2) \det(A^T) = \det(A)$$

$$(3) \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} = 0$$

$$T A = [v_1 \ v_2 \ \dots \ v_n]$$

$$A^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

$$(4) \det \begin{pmatrix} k \cdot v_1^T \\ v_2^T \end{pmatrix} = k \cdot \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix}$$

$$\det \begin{pmatrix} ka & kb \\ c & d \end{pmatrix} = k(ad - bc)$$

$$(5) \det \begin{pmatrix} v_2^T \\ v_1^T \end{pmatrix} = - \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix}$$

$$\det \begin{pmatrix} c & d \\ a & b \end{pmatrix} = bc - ad$$

$$(6) \det \begin{pmatrix} v_1^T + w_1^T \\ v_2^T \end{pmatrix} = \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} + \det \begin{pmatrix} w_1^T \\ v_2^T \end{pmatrix}$$

$$w_1 = \begin{bmatrix} e \\ f \end{bmatrix}$$

$$v_1 + w_1 = \begin{bmatrix} a+e \\ b+f \end{bmatrix} \quad \det \begin{pmatrix} a+e & b+f \\ c & d \end{pmatrix} = (a+e)d - (b+f)c = (ad - bc) + (ed - fc)$$

$$(7) \det \begin{pmatrix} v_1^T + kv_2^T \\ v_2^T \end{pmatrix}$$

$$\stackrel{(6)}{=} \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} + \det \begin{pmatrix} kv_2^T \\ v_2^T \end{pmatrix}$$

0 by (3) & (4)

$$= \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix}$$

These properties completely characterize
the determinant as a function on 2×2
matrices.